

SOO YAU MING

SOFTWARE ENGINEER | AI/ML DEVELOPER

Ipoh, Perak, Malaysia | [PHONE] | [PROFESSIONAL EMAIL] | [LINKEDIN] | [GITHUB]

PROFESSIONAL SUMMARY

Computer Science undergraduate and Research Assistant Intern with hands-on full-stack, AI/ML, mobile, and data engineering experience. Built role-based web systems, validated healthcare OCR workflows, and trained computer vision models using Next.js, React, TypeScript, Flutter, Python, Firebase, and cloud AI APIs.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Dart, SQL, HTML, CSS

Frameworks: Next.js, React, Flutter, Tailwind CSS, WeasyPrint

AI/ML: Computer Vision, OCR, CNNs, EfficientNet, Mask R-CNN, RAG, LLM Integration, Google ML Kit

Cloud & Tools: Firebase Authentication, Cloud Firestore, Supabase, BigQuery, OpenAI API, Gemini API, Git, REST APIs, OAuth 2.0, SSO

EXPERIENCE

Research Assistant Intern | Quest International University - Delta Lab | Apr 2026 - Present

Ipoh, Malaysia

- Delivered 40+ technical and UI improvements in one development cycle for the QIU Industry Day 2026 Vacancy Portal by implementing 4-tier role-based access control, Firebase authentication, degree-aware job matching, automated CV generation, employer approvals, CSV exports, analytics dashboards, and a grounded RAG assistant.
- Improved vacancy discovery and administrative auditability by building a 5-mode sorting engine, role-scoped dashboards, staged edit reviews, application tracking, and exportable records with Next.js, React, TypeScript, and Cloud Firestore.
- Expanded an aerial tree-classification pipeline to 135,569 images across 69,111 training, 13,523 validation, and 52,935 testing samples by partitioning the dataset and designing a two-stage Mask R-CNN plus EfficientNetV2-S architecture for 20 tree species.
- Raised 20-class validation accuracy by 1.81 percentage points, from 76.39% to 78.20%, by scaling EfficientNet-B3 training to 25 epochs; later reached 78.77% by migrating to EfficientNetV2-S and adding 25 augmented sample subsets.
- Validated a Flutter healthcare OCR pipeline against 48 synthetic blood reports covering 12 months for 4 patient profiles by creating a Python/WeasyPrint test-data generator and integrating Google ML Kit preprocessing with OpenAI API extraction.
- Launched a domain-restricted judging and voting portal after achieving 100% evaluation-panel submission compliance by auditing score matrices, processing participant media, enforcing OAuth/SSO access, deduplicating ballots, and applying deadline-based database locks.

LEADERSHIP & COMMUNICATION

- Co-facilitated an introductory machine learning practical by preparing annotated Python exercises and mentoring students on supervised learning, accuracy, precision, recall, and real-time debugging.
- Presented full-stack and AI prototypes to the Chief Operating Officer, Career and Professional Development Centre, lecturers, and research stakeholders, translating technical architecture into operational recommendations.

EDUCATION

Bachelor of Computer Science | Quest International University | Expected [MONTH YEAR]

Ipoh, Malaysia